.

INCH-POUND

.

A-A-59320
26 August 2003

COMMERCIAL ITEM DESCRIPTION
OHMMETER

The General Services Administration has authorized the use of this Commercial Item description (CID) for all federal agencies

1. SCOPE. This Commercial Item Description (CID) describes an ohmmeter. This CID is meant as a minimum requirement for the ohmmeter, in which only those manufacturers that meet or surpass the following requirements are supplied per this CID.

2. SALIENT CHARACTERISTICS. The equipment shall be capable of operation within the accuracies, limits, and specifications herein. The equipment shall be Class 3 in accordance with MIL-PRF-28800 except as specified herein.

2.1 Order of Precedence. In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications), specification sheets or MS standards the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

2.2 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

2.3 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 2000 hours of operation with a statistical certainty of 95%.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611

AMSC N/A FSC 6625

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:10Z
Check the source to verify that this is the current version before use.

2.4 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater, based on an operating time of 2000 hours per year.

2.5 Electrical Power Sources and Connections. The equipment shall operate from nominal commercial power source of 115/230 Volts AC (± 10 %) at line frequencies of 50 Hz to 60 Hz (± 5 %) or Internal Battery.

2.6 Internal DC Power. DC power shall be supplied by internal, rechargeable storage batteries. The equipment shall be capable of operating away from an external AC power source. The rechargeable batteries shall have the capability of operating the unit with full performance for a period of not less than four hours for the operating temperature range specified herein. The batteries shall conform to MIL-PRF-28800.

2.7 Recharge Time. The charger circuitry shall be such that a condition of charge, which shall permit at least four hours of continuous use of the unit, within all specification requirements, shall be reached in no more than sixteen hours following a low battery indication. Under no condition of discharge shall the charging rate applied to the battery exceed the battery manufacturer's recommended maximum rate, nor shall an indefinite charging period result in any damage to the battery or any other components.

2.8 Battery Charger. An internal battery charger shall be provided in accordance with MIL-PRF-28800, Section 3.

2.9 Accessories. Accessories to be supplied by the manufacturer with each deliverable piece of equipment shall include standard accessories as provided by the manufacturer and listed in the manufacturer’s catalog or data sheet, a written tutorial/user reference manual, a written programmer reference manual, a traceable calibration certificate, and a complete set of test leads with provisions for storage.

2.10 Temperature Requirement. Unless otherwise specified herein, the equipment shall meet its specified performance characteristics and other requirements when subjected to the class 3 temperature requirements specified by MIL-PRF-28800 and as listed below.

2.11 Non-Operating Temperature. The equipment shall meet its performance and accuracy requirements in a non-operating environment of – 10 ºC to + 50 ºC.

2.12 Operating Temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of 10 ºC to + 40 ºC.

2.13 Performance Characteristics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:10Z
Check the source to verify that this is the current version before use.

2.13.1 Resistance Ranges. The equipment shall have the capability to perform measurements from 0 to 199.99 ohms with sufficient range selection to meet the resolution and accuracy stated herein. The equipment shall have the capability to provide both automatic and manual range selection.

2.13.2 Accuracy. The accuracy of the equipment shall be at least ±0.2% of the reading plus three counts.

2.13.3 Maximum Test Current. The maximum continuity test current shall not provide more than 10 Amps for the ranges from 0 to 199.99 milliohms, and shall not provide more than 1 Amp for the ranges from greater than 199.99 milliohms to 199.99 ohms.

2.13.4 Maximum Open Terminal Voltage. The open terminal voltage shall not be greater than 2.0 Volts on any range.

2.13.5 Over-range. The equipment shall be provided a method for indicating an over-range condition.

2.13.6 Measurement Method. The equipment shall have a four terminal measurement method to assure measurement accuracy.

2.13.7 Display. The equipment shall have a display with the resolution to produce the required accuracy stated herein.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE

4.1 Product Conformance. The Products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerance using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

4.3 The contracting officer has the option of accepting or rejecting the product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:10Z
Check the source to verify that this is the current version before use.

5. PACKAGING.

5.1 Preservation, Packing, and Marking. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Sources of Documents.

6.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.1.2 The Code of Federal Regulation (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.2 Key Words: milli-ohms

MILITARY INTEREST

Custodian:
Air Force – 99

Preparing activity:
Air Force - 84

Agent:
Air Force - 99

(Project 6625-0927)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:10Z
Check the source to verify that this is the current version before use.